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Autonomous Mobile Robot for Warehouse Logistics

Collection Mission Planning and Motion Control
with Pedestrian-Aware Navigation in MATLAB

Francesco De Tursi

MSc student in Mechanical Engineering

University of Bologna

LinkedIn | GitHub | ffrancesco.detursi@gmail.com

Project overview

A personal project developed in MATLAB R2025b to simulate autonomous material collection in a warehouse. A single robot serves selected racks, completes timed loading and unloading operations, and returns to a waiting station. The application combines visit-order optimisation, path planning, motion control and pedestrian interaction, with recorded missions used to evaluate travelled distance, completion time and geometric separation.

Abstract

A single-robot warehouse collection simulator was developed using MATLAB R2025b. The robot receives a set of storage racks to serve, collects material through timed loading operations, visits an unloading station and returns to a fixed waiting station. The user can prescribe the visit sequence or request its optimisation. The project combines mission management, geometric path planning, smooth motion references, feedback tracking, pedestrian supervision and persistent-obstacle replanning within a reproducible software application.

The environment is a known two-dimensional warehouse containing six racks. A rectangular robot footprint is considered explicitly in geometric checks, while a circumscribed circle provides a conservative planning approximation. Station-to-station paths are generated by grid-based A* search, simplified and smoothed. Exact enumeration selects the minimum-cost visit order for the precomputed directed route matrix. Pedestrians move between randomly selected destinations using seeded local random streams; the robot receives ideal visible positions and velocities rather than future pedestrian intentions.

Nine recorded missions assess the complete collection cycle, visit-order optimisation and operation with three or five pedestrians. For the four-rack comparison without pedestrians, the travelled distance decreases from 91.21 to 69.86 m, a reduction of 23.41%, while total simulated duration decreases by 7.06%. All nine missions complete their requested loading operations, unloading and return. Across the six pedestrian runs, mean duration is 300.16 s with three pedestrians and 340.03 s with five. These averages describe the selected scenarios and are not population-level reliability estimates.

The results demonstrate an integrated, inspectable simulation of warehouse mission execution. They do not establish physical robot accuracy, industrial throughput or safety conformity. Ideal localisation, simplified pedestrian behaviour, geometric perception and the absence of vehicle and load dynamics define the main limits of the study.

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Nomenclature

Symbol or term	Meaning
AGV	Automated guided vehicle; used here for the simulated warehouse vehicle.
AMR	Autonomous mobile robot.
W, D	Waiting station and unloading station.
R01–R06	Storage racks.
P01–P06	Loading docks associated with the racks.
$\boldsymbol{q} = (x, y, \theta)$	Planar robot pose.
v, ω	Linear speed and yaw rate.
r_b	Radius of the robot's circumscribed circle.
r_h	Simulated pedestrian radius.
m_s, d_h	Static planning margin and required pedestrian separation.
C_{ij}	Length of the precomputed directed route from station i to j.
π	Permutation of the selected loading stations.
s, κ	Path arc length and curvature.
α	Supervisor speed/reference-clock scaling factor.
T, L	Total simulated mission duration and travelled distance.
M01–M09	Recorded evaluation missions.

Bold mathematical symbols denote vectors. Distances, speeds and times use SI units. Headings and angular rates use radians unless degrees are explicitly stated. A *hold* is a stationary supervisor state; it does not include every second spent slowing or restarting. A *route update* is an accepted replacement reference, which need not be geometrically different from an earlier accepted reference.

Chapter 1

Introduction and Project Objectives

1.1 Industrial context and motivation

Material movement connects storage, assembly, inspection and packaging operations. A useful transport task is therefore more specific than moving between arbitrary points: the vehicle must visit the correct locations, respect an operational sequence, stop for service and finish in a defined state. The present project models this collection cycle in a compact warehouse and investigates how route selection and pedestrian encounters affect its execution.

The contribution is the integration of mission selection, geometric planning, motion execution and pedestrian supervision. Separating these functions makes their decisions inspectable and connects an operational request to explicit constraints and recorded outcomes. The project demonstrates numerical modelling and automation methods within a simulation; it is not a production-ready vehicle controller.

1.2 Collection task and functional requirements

A mission begins at station W. The user selects a non-empty subset of six storage racks. Each rack has a corresponding loading dock where the robot remains stationary for a prescribed service time. After all selected docks have been served, the robot travels to D, waits for unloading and returns to W. No manipulator, lifting mechanism or loading operator is simulated. The service interval represents completion of an external material-transfer operation.

The functional requirements are summarised in Table 1.1. They provide the link between the operational purpose and the evaluation cases. In particular, a successful mission requires completion of the entire cycle, not merely arrival at the last rack.

1.3 Objectives and evaluation questions

Three questions guide the evaluation: does the robot complete every service and return to W; how much distance does ordering save for the specified four-rack request; and how do pedestrian encounters affect duration, stationary waiting and route updates?

The ordering comparison excludes pedestrians to isolate the effect of visit sequence. Traffic runs retain the optimised sequence and vary pedestrian configuration.

Table 1.1: Functional requirements of the simulated system.

ID	Requirement
F1	Select racks interactively and retain an explicit service sequence.
F2	Support manual ordering and distance-based order optimisation.
F3	Account for robot dimensions when planning and checking motion.
F4	Execute stationary loading and unloading before continuing.
F5	Slow, wait and resume in response to visible pedestrian conflicts.
F6	Attempt a new route when a stationary obstruction persists.
F7	Save the mission configuration, trajectory, events and performance data.
F8	Report incomplete execution explicitly instead of declaring false completion.

1.4 Scope of the implementation

The software uses base MATLAB R2025b. Its motion model is planar and kinematic, with ideal pose knowledge and a known static map. It does not require ROS, Simulink or a robotics toolbox. This choice keeps the numerical methods and their data flow directly accessible and reduces deployment dependencies for reproduction of the simulation.

The term AGV is used as a convenient description of the warehouse vehicle, although its software plans routes freely in a known map rather than following a physical guide. This naming choice does not assign an industrial product category or imply compliance. ISO 3691-4:2023 addresses driverless industrial trucks and their systems and includes both AGVs and AMRs within its stated examples [4].

The robot has no battery model, payload capacity limit, tyre slip, actuator lag or load-dependent braking model. Pedestrians are simulated agents, not measured human trajectories. Accordingly, the study evaluates the implemented algorithms under declared assumptions. Physical deployment and wider validation are discussed in Chapter 8.

Chapter 2

Warehouse Environment and Robot Model

2.1 Layout and coordinate system

Warehouse A occupies a rectangular area of 24×18 m. A fixed Cartesian frame places the origin at the lower-left corner of the map, with x increasing to the right and y upwards. Six rectangular storage racks are represented as occupied areas. Each rack is 2 m wide and 3 m deep in this plan view. Their lower-left corners are $(5,4)$, $(5,11)$, $(11,4)$, $(11,11)$, $(17,4)$ and $(17,11)$ m.

The arrangement creates longitudinal aisles and cross-aisle connections while keeping the map compact enough for visual inspection. It is a synthetic layout rather than a survey of an existing facility. Its purpose is to provide multiple station connections and encounters without hiding the behaviour in a very large environment. Figure 2.1 identifies the racks and operating stations.

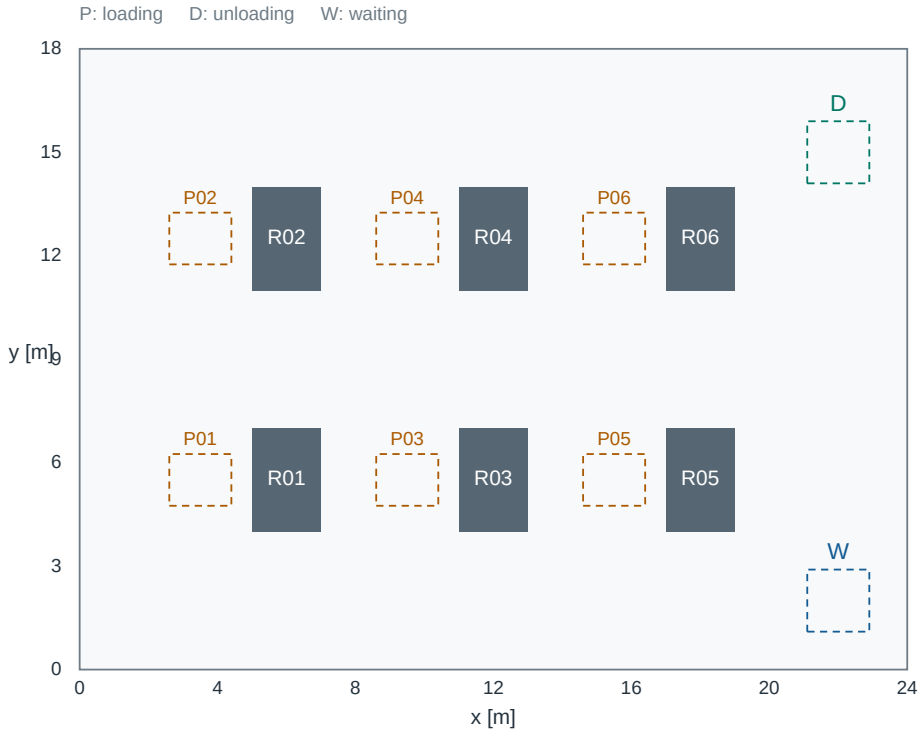


Figure 2.1: Warehouse geometry used by all nine missions. Dashed station rectangles denote operating zones; they are not additional physical obstacles.

2.2 Loading, unloading and waiting stations

Each loading pose is located 1.5 m to the left of the corresponding rack's left edge and at its vertical centre. The required loading heading is zero, so the vehicle faces the rack. The loading approach point lies 1.25 m behind that pose. This extra point gives path planning a consistent direction of entry and avoids treating service as unrestricted arrival anywhere near the rack.

The unloading pose is $(22, 15, \pi/2)$, with approach point $(22, 13.5)$ m. The waiting pose is $(22, 2, 0)$, approached from $(20.5, 2)$ m on return. Loading operating zones are 1.8×1.5 m; W and D use 1.8×1.8 m zones. Their graphical boundaries identify operating areas but do not constrain pedestrian motion as reserved resources. In particular, a loading zone can be occupied by a pedestrian and must then be handled by the same supervision logic as the rest of the map.

The final straight approach can be affected by smoothing near its upstream junction. The destination position and requested final heading remain explicit constraints. The robot may rotate in place after translational arrival to achieve the required service orientation. This is an important distinction from a continuously oriented docking controller for a physical loading interface.

2.3 Robot footprint and kinematics

The robot is represented by a rigid rectangle of length $\ell = 1.00$ m and width $b = 0.70$ m. Its state is $\boldsymbol{q} = (x, y, \theta)$, measured at the footprint centre. The commanded linear and angular velocities follow the unicycle equations

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega. \tag{2.1}$$

This captures forward motion and rotation in place, as could be achieved by a differential-drive platform. Wheel radii, wheel separation and individual wheel dynamics are not identified or simulated. The model therefore represents body motion rather than a detailed drivetrain.

For route planning, the rectangle is enclosed by a circle of radius

$$r_b = \tfrac{1}{2}\sqrt{\ell^2 + b^2} = 0.6103 \text{ m}. \tag{2.2}$$

Adding the static margin $m_s = 0.15$ m gives a planning clearance radius of approximately 0.7603 m. The circle avoids having to search over heading as well as position and protects against orientation changes along a path. Its cost is conservatism: a narrow passage that could accommodate the oriented rectangle may be rejected by the circular approximation. This follows the configuration-space principle of representing finite robot geometry through forbidden reference-point positions [2].

The executed trajectory is checked with the oriented rectangular footprint, not merely with the planning circle. Boundary distance and separation from every rack are evaluated. The planning approximation and execution check therefore serve different roles: one simplifies route generation; the other evaluates the realised pose sequence.

2.4 Motion limits and service assumptions

Table 2.1 lists the main execution settings. These are declared design assumptions for the study. They were not obtained from a particular commercial AGV or prescribed by a standard.

Table 2.1: Main motion and service settings.

Quantity	Value	Unit
Maximum commanded linear speed	0.80	m/s
Linear acceleration limit	0.40	m/s^2
Linear deceleration magnitude	0.60	m/s^2
Maximum commanded yaw rate	0.70	rad/s
Yaw acceleration magnitude	1.00	rad/s^2
Nominal simulation step	0.02	s
Docking reference speed cap	0.20	m/s
Docking reference distance	0.80	m
Lateral reference acceleration cap	0.30	m/s^2
Loading time per rack	8	s
Unloading time	12	s
Position tolerance at service	0.025	m
Heading tolerance at service	1	degree

A lower docking reference speed is appropriate because the objective changes from covering distance to reaching a service pose. However, the model does not represent contact alignment, pallet tolerances or an operator handover procedure. The 8 s loading and 12 s unloading values make service time explicit and reproducible; they are not measured production-cycle times.

Payload accumulation does not alter the footprint, motion limits or vehicle dynamics. Each selected rack is visited once, and every visit is treated as a completed loading operation after its dwell interval. Capacity constraints, item quantities and material compatibility are outside the present mission definition.

Chapter 3

System Architecture and Mission Management

3.1 Functional decomposition

The application separates configuration, map definition, mission selection, route planning, execution, pedestrian behaviour and presentation. Figure 3.1 summarises the main information flow. The decomposition permits changes in visual presentation without changing the motion algorithms, and allows recorded numerical data to be analysed independently of animation speed.

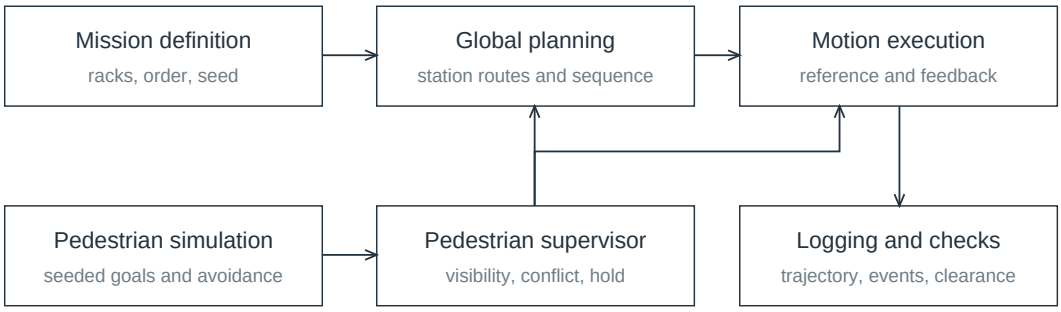


Figure 3.1: Functional architecture. The pedestrian supervisor affects execution at each simulation step and can request replacement of the current route.

The static map and robot settings are inputs to station-route planning. A mission request selects a subset of the loading docks and an ordering mode. The resulting plan contains the station sequence, directed legs and concatenated geometric path. Execution converts the path into motion references, applies tracking and supervision, and appends trajectory samples and discrete events. The same recorded run is subsequently used for replay and figure export.

The application computes the simulation before opening the replay. It is therefore an offline simulation with stepwise feedback, not a wall-clock real-time control process. The controller responds to the evolving simulated state at each step, but changing replay

speed has no effect on the computed mission. Computation time on the host PC must not be confused with the simulated mission duration reported in the results.

3.2 Interactive mission definition

The user selects racks directly on the map. The adjacent list displays the selected set and allows drag-and-drop reordering in manual mode. Clear-all and clear-selected controls avoid ambiguity when modifying a request. In optimised mode, the user still chooses the racks but delegates their order to the route-cost calculation.

Planning and execution are separate actions. *Plan mission* computes the complete W–loading–D–W cycle and displays its expected distance. *Run mission* executes that plan with the selected traffic configuration. Separating these actions allows the user to inspect the service sequence before committing to the simulation. The final interface offers no pedestrians or continuous random pedestrian traffic, with an explicit count and seed.

The mission definition contains no arbitrary start selection: W is both the initial and final station. This removes a degree of freedom that is not needed for the collection task and makes comparisons between requests easier. A revised start policy could be added for a fleet or dispatch system, but would need a corresponding change in the optimisation objective and evaluation conditions.

3.3 Execution states and service accounting

Each leg begins from the previous station and ends at the target service pose. Large heading changes between path segments can lead to an in-place rotation followed by translation. The controller tracks the reference, reduces speed near section ends and aligns to the requested final station heading. Arrival is accepted only when position, heading and near-zero motion satisfy the docking checks.

A loading event follows arrival at P01–P06. The completed-pick count increases only after the full loading dwell has elapsed. At D, unloading starts only after every selected loading station has been served. Return to W is recorded after the final leg. This event logic prevents a partially completed route from being interpreted as a finished collection mission.

Pedestrian supervision overlays the motion states. The log can show slowing, stationary hold and resumption, in addition to travel, rotation, docking, loading and unloading. The distinction is needed because time spent reducing speed is still part of execution, while the hold metric counts only samples explicitly labelled as a stationary hold. Service time is accounted for separately.

A random-traffic mission exceeding 1200 s is marked blocked and retains its partial records. A geometry assertion failure instead rejects the run as an error. Thus, incomplete progress and a violated execution condition are distinct outcomes. Neither occurred in the nine evaluation missions.

Chapter 4

Mission and Path Planning

4.1 Free-space representation and path search

The planner works with a known map of rectangular racks and warehouse boundaries. Candidate robot-centre positions are admitted only when their clearance exceeds the circumscribed robot radius plus static margin. Straight candidate segments are sampled, with an additional half-sample-step padding to cover positions between the samples. This deliberately conservative representation reduces the possibility of accepting a path that passes between valid endpoints but crosses an obstacle.

A grid spacing of 0.25 m defines the A* search graph. Each free grid point can connect to its eight neighbours, provided that the connecting segment is also free. Edge costs are Euclidean segment lengths. The start and goal are connected to nearby visible free grid points rather than assumed to coincide exactly with grid nodes.

The search priority combines accumulated cost and a Euclidean heuristic,

$$f(n) = g(n) + h(n), \qquad h(n) = \|\mathbf{p}_n - \mathbf{p}_g\|_2. \tag{4.1}$$

A* uses heuristic information to guide minimum-cost graph search [1]. Here, Euclidean distance is consistent with Euclidean edge costs. The graph solution is nevertheless restricted by the grid, the selected endpoint connections and obstacle inflation. Subsequent simplification and smoothing also change the geometry. The final continuous route is therefore not claimed to be the globally shortest path in the warehouse.

The grid approach was selected because it is deterministic, straightforward to inspect and sufficient for a small known map. Its resolution trades search effort against geometric detail. No resolution-convergence study was performed; 0.25 m is the implemented setting rather than an experimentally established optimum.

4.2 Simplification and smoothing

The raw graph path contains turns imposed by grid connectivity. A line-of-sight pass removes intermediate points when a longer straight connection remains admissible. This reduces unnecessary grid zigzags before a continuous transition is attempted at each remaining corner.

Corner smoothing uses a fifth-degree Bézier curve. If $\mathbf{P}_0, \ldots, \mathbf{P}_5$ are the control points and $u \in [0,1]$, the curve is

$$\mathbf{B}(u) = \sum_{j=0}^{5} \binom{5}{j} (1-u)^{5-j} u^j \mathbf{P}_j. \tag{4.2}$$

The endpoint control points lie on the incoming and outgoing segments, and their neighbouring control points are aligned with those segments. This preserves tangent directions at the connection. The initial trim length is limited by 1.4 m and 42% of each adjacent segment length, preventing the transition from consuming an entire short segment.

The candidate curve is checked against the inflated map. If it does not fit, the trim is halved and retried, up to seven attempts. When no acceptable curve is found, the original corner is retained. This fallback prioritises traversability over cosmetic smoothness. The method does not promise a globally curvature-continuous path, nor does it optimise curvature or jerk.

Straight portions and accepted curves are sampled at a nominal spatial scale of 0.025 m. These samples support reference construction and collision checks; they are distinct from the 0.25 m search grid and the nominal 0.02 s execution step. Keeping these three discretisations separate avoids attributing time-integration accuracy to map resolution.

4.3 Station routes and directional costs

A route to a station is planned through its approach point and then to its service pose. The final heading is enforced during execution. This directional arrival condition means that a route from i to j need not be equivalent to its reverse. The implementation therefore stores a directed cost matrix rather than assuming symmetry.

For an admissible sampled route $\mathbf{p}_{ij,0}, \ldots, \mathbf{p}_{ij,N}$, its cost is

$$C_{ij} = \sum_{k=1}^{N} \|\mathbf{p}_{ij,k} - \mathbf{p}_{ij,k-1}\|_2. \tag{4.3}$$

An unreachable connection receives infinite cost. Routes are cached while the map and robot configuration remain unchanged, allowing visit-order selection to reuse the same geometric calculations. Station service times do not appear in C_{ij}, and in-place orientation changes have no translational length. Thus, this cost is a distance criterion, not a prediction of total mission time.

The separation between route generation and sequence selection makes the optimisation tractable and transparent. It also fixes its scope: an optimised permutation is optimal for the stored matrix, not for every route that a different planner might generate.

4.4 Exact visit-order selection

Let $\pi = (\pi_1, \ldots, \pi_n)$ denote a permutation of the selected loading stations. The ordering objective is

$$J(\pi) = C_{W,\pi_1} + \sum_{k=1}^{n-1} C_{\pi_k,\pi_{k+1}} + C_{\pi_n,D} + C_{D,W}. \tag{4.4}$$

The final return cost is independent of the rack permutation, but remains in the reported total. In manual mode, π is the user's sequence. In optimised mode, the implementation enumerates all permutations and retains the lowest finite value of J.

There are 4! = 24 candidates for the main four-rack evaluation and 6! = 720 for the complete layout. Exact enumeration is reasonable at this scale and avoids introducing a heuristic whose quality would itself require evaluation. The helper limits requests to eight visits; the current map contains only six. Factorial growth makes this approach unsuitable for large warehouse orders without modification.

Deterministic candidate sorting and a numerical improvement tolerance give repeatable tie handling. The result is exact for Equation 4.4, not for routing with traffic, time windows or payload constraints.

4.5 Persistent-obstacle replanning

During execution, a prolonged hold may indicate that waiting alone will not restore progress. After 8 s of stationary conflict, the replanner considers currently visible pedestrians whose speed is at most 0.15 m/s. Only their current positions and velocities are used. Their future destinations and hidden positions are not supplied to this decision.

Selected detections are represented by temporary square obstacles. Their half-width is

$$p_h = r_h + \max(0, d_h - m_s) + 0.20, \tag{4.5}$$

which equals 0.65 m for the implemented parameters. Normal robot inflation is then applied by the route planner. The extra 0.20 m is a design allowance for a conservative detour around the snapshot, not a statistically calibrated uncertainty bound.

A new A* route is searched from the stopped robot to the current station's approach point. The approach-to-service segment must also be free. The route is simplified and smoothed using the same static checks. If departure space or station access is occupied, or no route is found, execution retains its previous reference and waits. A later clear observation can allow that reference to resume.

Accepted replanning requires zero linear and angular speed and does not alter the remaining station order. Attempts are separated by at least 12 s, with a maximum of 30 attempts per mission. These limits prevent unbounded repeated computation, but they do not prove freedom from deadlock. A snapshot route can become obstructed again as people move, so supervision remains active throughout its execution.

Accepted replacements can repeat an earlier route. Their count therefore measures replanning activity, not distinct detours, as illustrated by M07 in Chapter 7.

Chapter 5

Motion Execution and Pedestrian Interaction

5.1 From geometry to a timed reference

The reference generator estimates arc length, tangent heading and curvature from the sampled path. Speed is constrained by translational, lateral-acceleration and yaw-rate limits, with an allowance for curvature variation. For non-zero curvature,

$$v_r \leq \sqrt{a_{\text{lat}}/|\kappa|}. \tag{5.1}$$

Nominal cruising speed is capped at $0.95\,v_{\max}$ and the curvature-based yaw-rate allowance at $0.8\omega_{\max}$, leaving headroom for feedback.

Forward and backward passes enforce acceleration and braking constraints. The forward condition is

$$v_{k+1}^2 \leq v_k^2 + 2a_k\Delta s_k. \tag{5.2}$$

The backward pass ensures sufficient braking distance. Curvature-dependent allowances and time dilation limit yaw-rate changes. Segment lengths and adjacent speeds define the time coordinate for interpolation. The method does not optimise travel time or jerk.

Sections begin and end at rest. Two-point sections use triangular or trapezoidal profiles. Heading discontinuities greater than 12° split the route and can require in-place rotation; loading, unloading and final waiting also require stops.

The last 0.8 m of an approach has a reference-speed cap of 0.2 m/s. Position and heading are checked before service begins. These are design settings, not limits prescribed by a standard.

5.2 Tracking and numerical integration

The controller compares the current pose with a reference pose. Translational errors are expressed in the robot frame:

$$\begin{bmatrix} e_x \\ e_y \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_r - x \\ y_r - y \end{bmatrix}, \qquad e_\theta = \operatorname{wrap}(\theta_r - \theta). \tag{5.3}$$

Desired body commands are calculated as

$$v_d = \alpha\big(v_r \cos e_\theta + k_x e_x\big), \tag{5.4}$$

$$\omega_d = \alpha\big(\omega_r + k_y v_r e_y + k_\theta \sin e_\theta\big), \tag{5.5}$$

with gains $k_x = 1.5$, $k_y = 4$ and $k_\theta = 3$ in the consistent SI formulation. Saturation and rate limits then restrict the actual commands. The factor $\alpha \in [0,1]$ is used by pedestrian supervision; without a conflict, it tends to one.

The reference clock advances according to the average scaling factor over the step,

$$\Delta\tau = \tfrac{1}{2}(\alpha_k + \alpha_{k+1})\Delta t. \tag{5.6}$$

Clock scaling keeps the reference aligned with the robot during slowing and waiting. At full hold both are stationary; command acceleration and deceleration remain bounded during transitions.

Pose integration uses mean step velocities and a constant-curvature arc update, including its straight-line limit. With no actuator lag, slip or localisation uncertainty, tracking errors quantify numerical consistency rather than hardware accuracy.

5.3 Continuous seeded pedestrian traffic

Pedestrians are discs of radius 0.30 m, moving repeatedly between destinations on a 20-node circulation graph. Its columns are $x = 3, 9, 15, 21$ m and rows $y = 2, 5.5, 8.8, 12.5, 16$ m. Vertical edges connect adjacent nodes in each column; horizontal edges connect the lower, middle and upper cross-aisles.

Initial nodes are distinct and remain away from the robot's waiting pose. Preferred walking speeds are drawn between 0.65 and 1.00 m/s. After reaching a destination, a pedestrian pauses for a seeded interval of 1–5 s and chooses another destination. Initial pauses can be shorter. The destination sequence is reproducible for a fixed configuration, but is not supplied to the robot controller.

Local movement is not unconstrained random motion. Each pedestrian evaluates forward and alternative headings, full or half preferred speed, and a stationary option. A 1.5 s prediction considers other pedestrians and the robot; obstacle checks reject candidates entering racks. Selection balances goal progress with movement continuity and a penalty for remaining stationary. The nominal pedestrian acceleration limit is 0.8 m/s^2.

After more than 6 s of poor progress, a pedestrian can use a local grid search to detour around the robot. This crowd-recovery mechanism is separate from AGV replanning and its recorded update count.

Pedestrians use current robot geometry to avoid it, making separation a joint outcome of cooperative crowd motion and robot supervision. Sequential updates can introduce ordering bias. The behaviour is not calibrated against human observations and does not cover inattentive or adversarial movement.

5.4 Visibility, anticipation and restart

The perception abstraction provides pedestrian centres and velocities within an 8 m radius and with unobstructed line of sight from the robot centre. Rectangular racks occlude this line. The view is otherwise ideal and covers all directions. Laser rays, reflection properties, scanner mounting, measurement noise and missed detections are not simulated. This implementation is therefore not a validated LiDAR model.

For visible pedestrians, the supervisor extrapolates current velocity over 3 s in 0.2 s increments. It compares these positions with points on the upcoming robot reference over 5 s, sampled every 0.3 s. A conflict is identified when these spatial sets come sufficiently close. The threshold includes robot radius, pedestrian radius, required separation and an additional 0.15 m anticipation allowance.

The test compares spatial sets without requiring simultaneous arrival at a point. It is therefore conservative rather than an exact time-to-collision calculation, and can trigger unnecessary waiting.

On conflict, the scaling factor decreases, producing a controlled slowdown and, when necessary, a stationary hold. A clear observation must persist for 1 s before release; scaling then recovers at a bounded rate of 0.25 s^{-1}. This hysteresis reduces rapid switching between stop and go. A persistent stationary obstruction can invoke the replanning procedure in Chapter 4.

5.5 Separation checks and their meaning

The geometric checker calculates distance from each pedestrian centre to the oriented robot rectangle, then subtracts the pedestrian radius. Static clearance is evaluated against racks and the map boundary. These quantities refer to body surfaces rather than centre-to-centre distance.

Sampling alone could miss a smaller gap between recorded poses. The code also evaluates a conservative interval estimate. With endpoint gaps g_k, g_{k+1}, robot travel increment Δs_b, heading change $\Delta\theta$ and the largest pedestrian displacement magnitude $\|\Delta\mathbf{p}_h\|$, the pedestrian estimate is

$$\underline{g}_k = \min(g_k, g_{k+1}) - \tfrac{1}{2}\left(\Delta s_b + r_b|\Delta\theta| + \|\Delta\mathbf{p}_h\|\right). \tag{5.7}$$

The static estimate omits the pedestrian term. This allowance bounds possible gap reduction under the implemented piecewise motion assumptions. It does not cover sensing errors or motion between samples that exceeds those assumptions.

Runtime checks require a 0.15 m static margin and 0.30 m pedestrian separation. They establish geometric consistency under the model assumptions; physical protective-system validation is discussed in Chapter 8.

Chapter 6

Evaluation Methodology

6.1 Evaluation questions and controlled conditions

The evaluation asks whether the application completes its collection cycle, whether visit-order optimisation reduces the requested route length, and how continuous pedestrian traffic affects progress and separation. The same warehouse, robot footprint, service times and motion settings are used throughout. Nine recorded missions provide a compact evaluation set rather than a large statistical campaign.

The first two runs isolate the ordering choice. Both collect from R02, R05, R04 and R03 without pedestrians. M01 follows that selected order; M02 lets the planner choose a permutation. Because the remaining settings and service operations are unchanged, their distance difference can be attributed to ordering within the implemented planner.

M03–M08 retain the same four-rack optimised request and introduce three or five pedestrians, with seeds 42, 7 and 2027. These runs exercise encounters, holds, restarts and occasional replanning. M09 serves all six racks without pedestrians, checking the full set of service poses and the complete collection cycle. Table 6.1 defines the cases before their outcomes are discussed.

Table 6.1: Recorded mission matrix. Optimised ordering uses the static directed route-cost objective.

ID	Ordering	People	Seed	Selected racks
M01	Manual	0	–	R02, R05, R04, R03
M02	Optimised	0	–	Same four racks
M03	Optimised	3	42	Same four racks
M04	Optimised	3	7	Same four racks
M05	Optimised	3	2027	Same four racks
M06	Optimised	5	42	Same four racks
M07	Optimised	5	7	Same four racks
M08	Optimised	5	2027	Same four racks
M09	Optimised	0	–	R01–R06

Local random streams separate traffic generation from MATLAB's global random state [3]. One stream sets initial placements and preferred speeds; each pedestrian has a separate seeded stream for subsequent choices. Reproduction requires the same

seed, pedestrian count, selected racks, ordering mode, map, motion parameters and implementation.

Changing crowd size changes the initial random draws and subsequent interactions. Equal seed identifiers at three and five pedestrians therefore do not imply matched trajectories. Group summaries are descriptive and are not used to infer a general relationship between crowd size and completion time.

6.2 Distance, time and completion metrics

Planned distance is the sum of stored route costs for the selected station sequence. Travelled distance accumulates the robot-centre arc length using mean step speeds:

$$L = \sum_{k=1}^{N} \frac{v_{k-1} + v_k}{2} (t_k - t_{k-1}). \tag{6.1}$$

The implemented linear speeds are non-negative. Planned and travelled distances can differ because of discretisation, tracking and accepted route replacement. Translational distance excludes in-place rotation; rotation still consumes mission time. The distance reduction of optimised relative to manual execution is

$$\eta_L = 100 \frac{L_{\text{manual}} - L_{\text{optimised}}}{L_{\text{manual}}}. \tag{6.2}$$

Percentages are calculated from unrounded CSV values, then rounded for presentation.

Mission time extends from initial motion processing at W to completion of the return. Its accounting identity is

$$T = T_{\text{motion}} + T_{\text{hold}} + T_{\text{load}} + T_{\text{unload}}. \tag{6.3}$$

Here T_{motion} includes travel, turning, docking and slowdown/restart transitions. The exported turning-time field is a subset of this term and must not be added again. Hold time counts only the explicitly stationary pedestrian-hold state. Pedestrian delay is therefore not generally equal to recorded hold time.

Four-rack missions contain 32 s of loading and 12 s of unloading. The six-rack case contains 48 s and 12 s, respectively. Completed status additionally requires every requested pick, unloading and return to W. A plausible path image alone would not establish successful service completion.

6.3 Geometry and controller diagnostics

Minimum sampled gap and the conservative interval estimate from Equation 5.7 are reported separately. The interval estimate is used when comparing a run with the configured margin. Positive sampled distance alone would provide weaker evidence because it considers only stored instants.

Tracking error measures separation between the robot centre and contemporaneous reference centre. It checks numerical execution, but depends on ideal localisation and

the kinematic plant. Final position error similarly describes the simulated return pose. Neither quantity is used to claim physical docking accuracy.

Replanning attempts, accepted updates and hold episodes describe supervisory activity. An accepted update need not shorten the remaining path or differ from a previous update. Pedestrian emergency-stop counts belong to the pedestrian model's geometry guards, not to a certified AGV emergency-stop system. This distinction is retained when discussing the zero counts in the recorded runs.

6.4 Traceability and evidence limits

A MAT record preserves each mission configuration, plan and structured results, including accepted replacement routes. CSV files expose trajectory samples, service events, metrics and, for traffic runs, pedestrian states, supervisor transitions and replanning attempts. Analysis cross-checks final trajectory time and distance against summary metrics, service events against the requested station sequence, and time components against Equation 6.3. All nine records are internally consistent within numerical precision.

Mission IDs M01–M09 identify the configurations in Table 6.1. Trajectory plots and numerical summaries are derived from the corresponding saved records. Percentages use unrounded values; presentation rounding does not enter the calculations.

The study contains no sensor-noise campaign, physical experiment, convergence study, energy measurement or large random-seed sweep. It establishes the reported outcomes for these configurations, not universal success, an industrial throughput guarantee or a validated safety performance level. These boundaries define the scope of the conclusions.

Chapter 7

Results and Discussion

7.1 Mission completion and overall results

All nine recorded missions completed their requested loading operations, unloaded at D and returned to W. This is nine successful cases out of nine evaluated, including six with continuous pedestrian traffic. It is a descriptive count for the selected scenarios, not an estimate of success probability in an arbitrary warehouse.

Table 7.1 reports the main outcomes. Planned distance is 91.2110 m for M01, 69.8576 m for M02–M08 and 80.0327 m for M09. Small differences between planned and executed values occur even without route replacement because execution and path-length calculations use different discretisations.

Table 7.1: Executed mission outcomes. Updates are accepted replacement references, not necessarily distinct detours.

Mission	Distance [m]	Time [s]	Hold [s]	Updates
M01	91.21	254.82	0.00	0
M02	69.86	236.83	0.00	0
M03	69.86	276.66	27.36	0
M04	69.86	315.66	49.20	0
M05	70.12	308.16	34.90	1
M06	71.34	379.96	84.56	1
M07	70.12	363.30	84.54	3
M08	69.86	276.84	21.44	0
M09	80.03	299.68	0.00	0

The smallest conservative static-clearance estimate across all runs is 0.4170 m, in M09, exceeding the configured 0.15 m margin. The six pedestrian runs have conservative robot–pedestrian gap estimates between 0.4405 and 0.4551 m, above the configured 0.30 m separation. No pedestrian emergency geometry guard was counted in these runs. These observations concern the specified idealised model and its interval checks.

7.2 Effect of visit-order optimisation

M01 follows the selected order R02–R05–R04–R03; M02 selects R05–R03–R02–R04 by minimising the directed route cost. Both missions start at W, serve the same four racks, unload at D and return to W, without pedestrians. Figure 7.1 compares their executed paths on identical map extents.

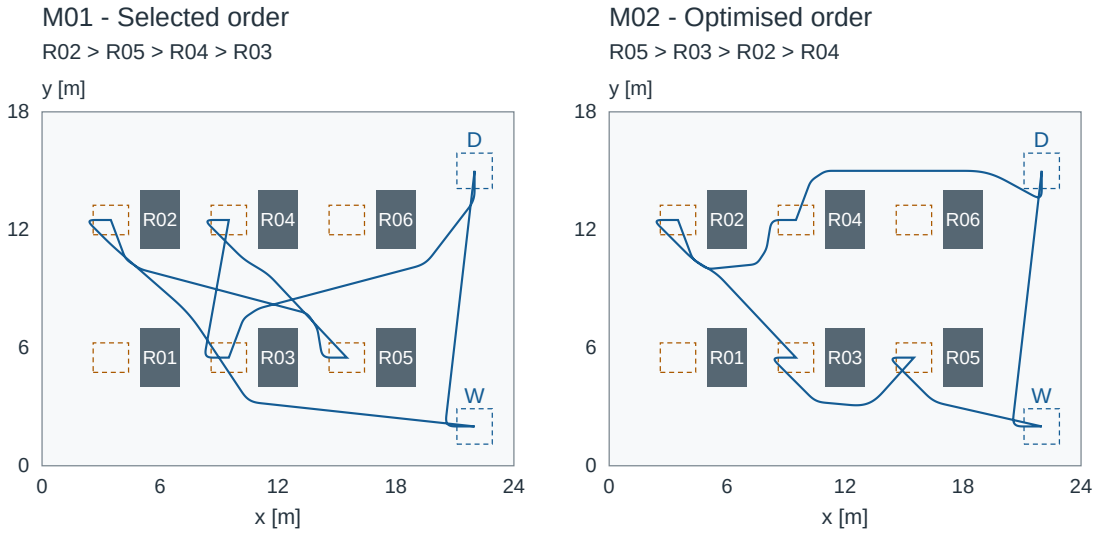


Figure 7.1: Executed paths for the manual and optimised missions. Blue denotes robot travel; amber boxes identify loading docks. Repeated aisle traversals can overlap in plan view.

Travelled distance decreases from 91.2105 to 69.8570 m: a saving of 21.3535 m, or 23.41%. The planned-distance reduction is also 23.41% at the reported precision. The different sequence connects the same service locations with less travel; it does not remove a loading operation or change the docking requirements.

Total duration decreases from 254.8247 to 236.8310 s, a saving of 17.9937 s or 7.06%. Both missions include 44 s of service. Turning, station approaches and acceleration constraints also consume time, so elapsed time does not scale directly with distance. The optimiser minimises route length, not completion time.

The benefit depends on the reference sequence: selecting the optimised order manually would produce no ordering gain. Energy consumption is not modelled, so the measured distance reduction cannot be converted directly into an energy saving.

7.3 Pedestrian traffic and mission time

Relative to the 236.83 s no-traffic mission, completion times range from 276.66 to 315.66 s with three pedestrians and from 276.84 to 379.96 s with five. Mean durations are 300.16 and 340.03 s, respectively. Figure 7.2 separates motion, stationary hold and service time.

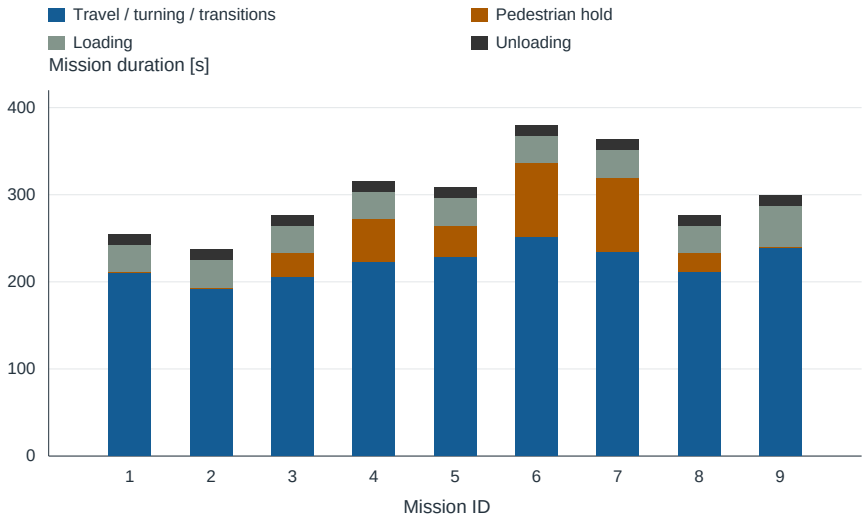


Figure 7.2: Time decomposition across the nine missions. Motion includes turning, docking and slowdown/restart transitions.

Mean stationary hold is 37.15 s with three pedestrians and 63.51 s with five. Nevertheless, M08 finishes sooner than M04 and M05: encounter timing matters as well as crowd size. Equal seeds at different counts do not imply matched pedestrian trajectories (Figure 7.3).

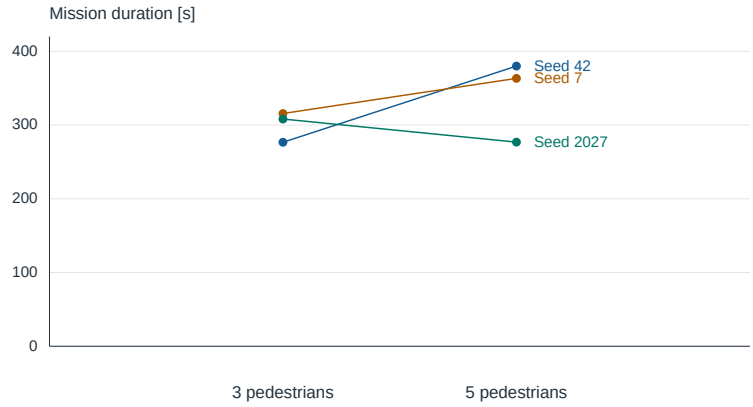


Figure 7.3: Individual completion times. Lines connect seed identifiers, not matched pedestrian trajectories.

M03 takes 39.83 s longer than M02 but records only 27.36 s of stationary hold; slowing and restarting explain the additional motion-time penalty. Distances remain close to the static plan: 71.34 m in M06 and about 70.12 m in M05 and M07. Traffic therefore affects duration more strongly than route length in these cases.

7.4 Detailed interaction case: M07

With five pedestrians and seed 7, M07 completes in 363.30 s, travels 70.1232 m and records 84.54 s of stationary hold over 16 episodes. Its speed and clearance histories are shown in Figure 7.4. Zero-speed intervals include service as well as traffic stops; the event log distinguishes their causes.

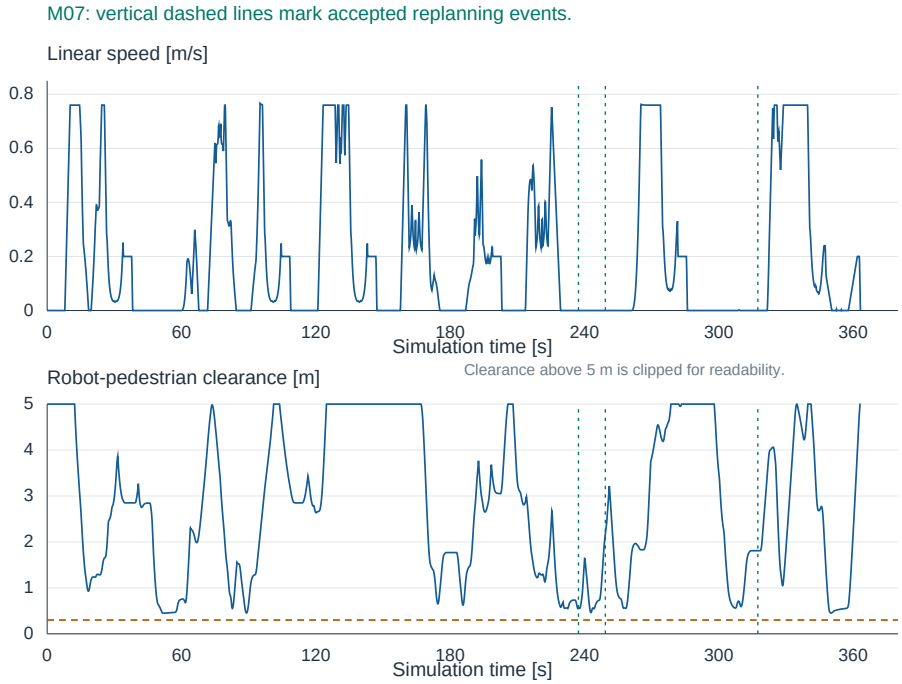


Figure 7.4: M07 speed and sampled pedestrian clearance, capped at 5 m for display. The horizontal dashed line marks the configured 0.30 m separation; vertical markers indicate accepted route updates.

Table 7.2: M07 replanning attempts. Length refers to the accepted replacement route to the current target.

Time [s]	Target	Outcome	Length [m]
183.38	P04	No stationary detection	–
237.32	D	Accepted	9.8657
249.32	D	Accepted	9.8657
317.44	W	Accepted	14.6452

The two equal-length D updates indicate repeated route acceptance after the cool-down, rather than two demonstrably distinct detours. The mission still completes its remaining service sequence. Its conservative pedestrian gap is 0.4414 m and maximum tracking error is approximately 4.47 mm. These are simulation diagnostics under ideal localisation, not measurements of physical docking accuracy.

7.5 All-rack collection and overall interpretation

M09 visits R05–R03–R01–R02–R04–R06, unloads and returns to W. It travels 80.0319 m and completes in 299.6753 s, including 48 s of loading and 12 s of unloading. Figure 7.5 shows all six loading poses incorporated in one request.

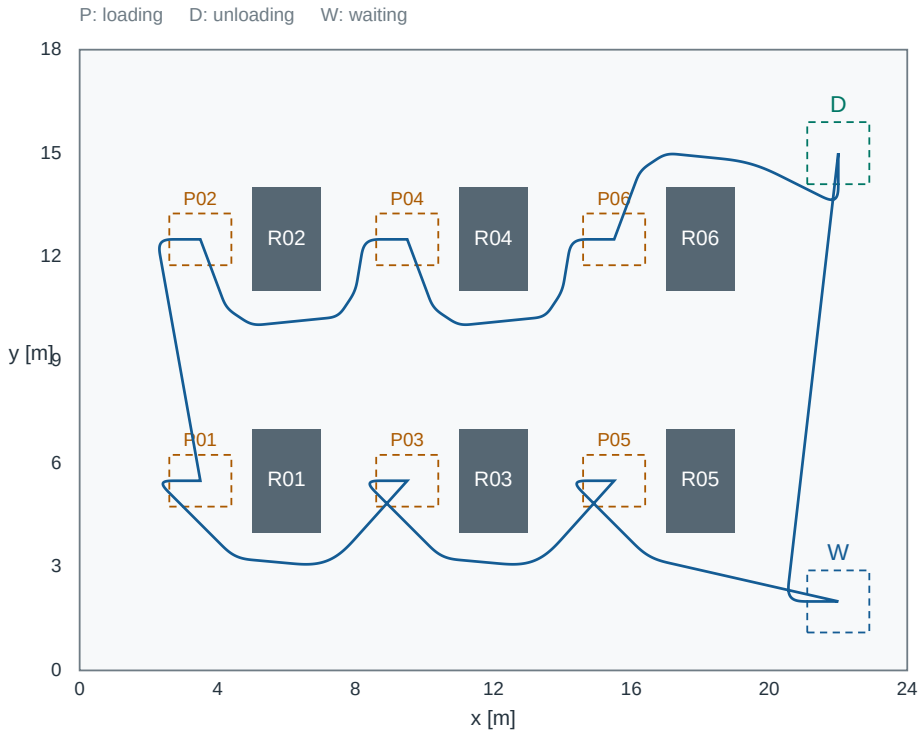


Figure 7.5: M09: executed collection route serving every rack, followed by unloading and return. This case contains no pedestrians.

The all-rack route is shorter than the four-rack manual route of M01. This is not contradictory: stop count alone does not determine length, and an unfavourable order can require backtracking. Its time is nevertheless longer than M02 because it adds two services and their approaches. Distance, service count and time must be interpreted together.

Together, the runs demonstrate end-to-end execution, a measurable ordering benefit, bounded geometric separation in the recorded interactions and recovery through waiting or occasional route replacement. They do not isolate the causal benefit of each supervisor component. In particular, there is no supervisor-disabled baseline or replanning ablation, so no time saving is attributed to replanning relative to an unobserved alternative.

Chapter 8

Industrial Relevance and Model Limitations

8.1 Connection to material-handling applications

The collection cycle represents a simplified material-handling task: an operator specifies storage locations, the vehicle visits them, transfers the collected load at a common station and returns to an available state. Similar planning questions arise in internal logistics, component collection and material supply around production equipment. The project is not tied to a particular manufacturer's warehouse or production line.

Its engineering value lies in combining responsibilities in one traceable application. Map geometry constrains feasible routes; station orientation affects arrival; order selection reduces cost; motion generation respects declared limits; supervision handles evolving traffic; and event records distinguish travel from service completion. These responsibilities are transferable to a more detailed implementation even though the present plant is simulated.

The simulation separates route efficiency from traffic delay. Distance-based ordering reduces unnecessary travel for a fixed collection request, while hold and completion times expose the effect of shared-space operation. This distinction is relevant when evaluating material supply cycles: a short route alone does not determine the time required to complete an order.

8.2 Safety references and deployment requirements

ISO 3691-4:2023 addresses safety requirements and verification for driverless industrial trucks and their systems [4]. ISO 12100:2010 provides general principles for machinery risk assessment and risk reduction [5]. The cited catalogue scopes establish the relevance of these standards to the application. The full standards were not used for a clause-by-clause assessment, and the simulation is not claimed to conform to either standard. Draft revisions are not treated as replacements for the cited published editions.

The connection is methodological: define the operating context and hazards, distinguish protective functions from ordinary navigation, and identify the verification needed for a real system. Choosing a small numerical separation threshold or completing simulations is insufficient to establish compliance.

A real protective field must account for the complete stopping process. SICK's mobile-scanner guidance separates braking distance from distances travelled during vehicle-control and scanner response [6]. This illustrates why the simulated 0.30 m separation must not be copied into a physical scanner configuration: the model has no measured sensor latency, load-dependent brake test or validated control-response time.

For illustration only, constant-deceleration braking from speed v after delay t_r gives

$$d_{\text{stop}} = v\,t_r + \frac{v^2}{2a_b}. \tag{8.1}$$

Even with zero delay, $v = 0.8$ m/s and $a_b = 0.6$ m/s^2 yield approximately 0.53 m before additional allowances. This is not a field-sizing rule; it shows that clearance, anticipated conflict distance and stopping distance are different quantities. The supervisor anticipates conflicts before reaching the configured minimum gap.

In this simulation, speed reduction on curves follows curvature, yaw-rate and acceleration limits; full stops follow docking, discontinuous heading changes or supervision. A physical vehicle's permitted motion would require its own load, stability, operating-area and protective-system assessment.

8.3 Limitations affecting interpretation

The known map and ideal localisation remove mapping and localisation uncertainty. Perception exposes exact centres and velocities for visible people, without measurement errors or classification. The work evaluates navigation and mission logic under an ideal observation model, rather than a complete autonomous sensing stack.

The kinematic plant omits traction, actuator bandwidth, drivetrain geometry and payload effects. Service is represented by elapsed time without a transfer mechanism or interlock. Capacity and item quantities are absent. These simplifications make mission logic testable but prevent conclusions about payload capacity, energy use or actual handling-cycle time.

Pedestrians are cooperative, graph-guided agents. They can avoid the robot and use privileged geometric information in their movement model. Results do not describe arbitrary human behaviour or an adversarial collision challenge. Occlusion is represented, but unexpected acceleration, perception failure and entry from unmapped areas are not systematically tested.

The nine missions provide useful case evidence but limited coverage. There is no large-sample reliability estimate, parameter sensitivity study or experiment separating the contributions of smoothing, anticipation and replanning. Exact visit-order enumeration is limited to small requests. These are concrete boundaries for future evaluation rather than reasons to infer unmeasured performance.

8.4 Priorities for further development

The first extension should strengthen validation while retaining the single-robot architecture. A larger fixed seed set, difficult encounters, blocked-dock cases and numerical step-size checks would establish a broader operating envelope. Recording rejection causes and distinguishing equivalent route updates would improve interpretation of replanning activity.

A second extension would introduce sensing and vehicle uncertainty: limited field of view, delayed and noisy observations, a localisation model and a differential-drive plant with actuator dynamics. Hardware experiments would then identify braking and tracking behaviour. Simulation can support preparation but cannot replace those measurements.

For larger orders, factorial enumeration should be replaced by a scalable routing method with a documented optimality gap or comparison against exact small cases. Capacity, priorities and time windows would turn the collection request into a richer operational problem. The objective would need revision rather than retaining distance as an implicit proxy for all costs.

Finally, extending to two or three vehicles requires task allocation, shared-dock access, corridor coordination and deadlock handling. Instantiating the current controller several times would not be sufficient. A fleet evaluation should compare completed orders per unit time, travel, congestion and resource occupancy across fleet sizes under the same demand. The present system is limited to one AGV.

Chapter 9

Conclusions

The project implements a complete simulated warehouse collection cycle in MATLAB: rack selection, optional visit-order optimisation, travel to oriented loading poses, timed loading, unloading and return to a waiting station. It combines finite-footprint planning, smoothed references, kinematic tracking and continuous seeded pedestrian traffic in an inspectable software structure.

All nine recorded evaluation missions completed. For the specified four-rack request without pedestrians, optimised ordering reduces travelled distance from 91.21 to 69.86 m, or 23.41%, relative to the selected manual order. Total simulated duration falls by 7.06%. The distinction is consistent with distance-based optimisation and fixed service operations.

The six pedestrian cases demonstrate slowing, holding, resumption and occasional accepted route replacement while retaining the collection sequence. Their conservative robot–pedestrian separation estimates remain above the configured 0.30 m value. Completion time varies substantially between seeds: route efficiency and traffic exposure are separate aspects of performance. Replanning counts describe accepted updates without assuming that every update is a distinct beneficial detour.

The evaluation verifies the integrated mission logic under the stated simulation assumptions. Ideal perception, cooperative pedestrians and kinematic motion limit its transfer to a physical vehicle. Hardware accuracy, protective-system performance and operational throughput require additional modelling and experimental validation.

The application provides a coherent foundation for broader validation, realistic sensing and dynamics, richer collection constraints and eventually coordinated multi-vehicle operation. The present contribution is a reproducible single-robot system with an operational purpose and results traceable to saved mission records.

References

[1] P. E. Hart, N. J. Nilsson and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968. Article on IEEE Xplore.

[2] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006. Author's online edition.

[3] MathWorks, "RandStream: Random number stream," MATLAB documentation. Official documentation. Accessed October 2026.

[4] International Organization for Standardization, *ISO 3691-4:2023: Industrial trucks—Safety requirements and verification—Part 4: Driverless industrial trucks and their systems*, second edition, June 2023. Official catalogue and scope. Accessed October 2026.

[5] International Organization for Standardization, *ISO 12100:2010: Safety of machinery—General principles for design—Risk assessment and risk reduction*, 2010. Official catalogue and scope. Accessed October 2026.

[6] SICK AG, *microScan3—PROFINET: Safety Laser Scanner, Operating Instructions*, document 8021219/1SK3, July 2025, mobile-application stopping-distance guidance, pp. 50–51. Official operating instructions. Accessed October 2026.

Appendix A

Parameter and Diagnostic Tables

The following settings complement the robot and service parameters in Table 2.1. They describe the evaluated implementation; none is presented as a value prescribed by a safety standard.

A.1 Planning and supervision settings

The geometric and supervisory parameters act at different stages: the grid controls search resolution, path sampling supports reference generation, and the temporal thresholds govern hold, restart and replanning.

Table A.1: Planning and supervision parameters.

Parameter	Value
Search grid spacing	0.25 m
Nominal geometric sampling spacing	0.025 m
Static planning margin	0.15 m
Initial corner trim limit	1.4 m; 42% of adjacent lengths
Corner-fitting attempts	7, with halved trim
Section heading-break threshold	12°
Maximum ordering-helper request	8 stations (6 available)
Required robot–pedestrian separation	0.30 m
Ideal pedestrian visibility radius	8 m
Pedestrian prediction horizon / step	3 s / 0.2 s
Robot reference look-ahead / step	5 s / 0.3 s
Additional anticipation allowance	0.15 m
Clear interval before restart	1 s
Supervisor scaling rate	0.25 s^{-1}
Stationary hold before replanning	8 s
Replanning attempt cooldown	12 s
Maximum replanning attempts	30
Stationary-detection speed threshold	0.15 m/s
Temporary pedestrian obstacle half-width	0.65 m
Random-traffic mission deadline	1200 s

A.2 Pedestrian settings and interaction diagnostics

The pedestrian parameters control the variability and responsiveness of simulated traffic. Preferred speed and pause duration are sampled within the stated ranges; local avoidance can reduce actual speed or introduce a stop.

Table A.2: Pedestrian motion and recovery settings.

Parameter	Value
Pedestrian radius	0.30 m
Preferred pedestrian speed range	0.65–1.00 m/s
Nominal pedestrian acceleration limit	0.8 m/s^2
Pedestrian local prediction horizon	1.5 s
Pause at a pedestrian destination	1–5 s
Pedestrian stalled-progress threshold	6 s
Pedestrian recovery grid spacing	0.5 m

Table A.3: Conservative robot–pedestrian gap and replanning diagnostics. Every run in this table completed.

Mission	Gap bound [m]	Attempts	Accepted updates
M03	0.4423	2	0
M04	0.4405	1	0
M05	0.4551	2	1
M06	0.4428	3	1
M07	0.4414	4	3
M08	0.4439	0	0

The gap column is the interval estimate, not simply the minimum sampled separation. Accepted replacements are counted according to the existing logger. The repeated equal-length updates in M07 remain in the reported count, as explained in Chapter 7. An accepted update is counted even when its length matches a previous update.